



JewelDesk

Jewellery Retail Management & ERP System

Complete Project Architecture, Feature Specifications & Operational Manual

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System Classification: Enterprise Retail Jewellery Management System

Technical Foundation: Django 6.1 • Python 3.12 • Bootstrap 5.3 • ReportLab • Progressive Web App (PWA)

Database: SQLite (Local Dev) / PostgreSQL (Production & Cloud Hosting)

EXECUTIVE SUMMARY:

JewelDesk is an end-to-end, production-grade jewellery showroom management, billing, inventory, and artisan (karigar) manufacturing ERP application. Specifically tailored to the Indian and global jewellery trade, the application incorporates statutory GST (3%) calculation, Hallmark Unique Identification (HUID) compliance, purity calculations (24K, 22K, 18K, 925), dynamic pricing based on live metal rates, customer old gold exchange and cash buybacks, bespoke custom order lifecycles, artisan metal loss/wastage tracking, role-based employee access control, and complete Progressive Web App (PWA) offline resilience.

System Core Modules Summary:

Module	Core Functionality	Key Business Benefits
1. Cockpit Dashboard	Executive summary, sales totals, inventory value, pending dues, live metal prices, alerts.	Instant business visibility and one-click actions.
2. Inventory & Stock	HUID tag tracking, purity, weight details, making charges, CSV import, audit movement log.	Zero stock leakage; full hallmark compliance.
3. Pricing Engine	Live gold & silver price feeds, dynamic formula calculation, live AJAX price calculator.	Accurate real-time retail pricing automatically.
4. Sales & Invoicing	Multi-item billing, GST (3%), split payments, ReportLab PDF tax invoices, WhatsApp billing.	Instant retail billing with print & mobile sharing.
5. Old Gold Exchange	Karat testing, melt value calculation, wastage deduction, invoice offset or cash buyback.	Profitable and transparent scrap metal handling.
6. Customer CRM	Customer directory, purchase ledger, PAN/Aadhaar compliance, enquiries and lead follow-up.	Repeat sales and statutory compliance tracking.
7. Custom Orders	Bespoke design specs, reference photo uploads, advance payments, 8-stage progress tracking.	Eliminates delivery delays and design mistakes.
8. Karigar & Artisans	Raw metal issue, work order assignments, scrap return, metal loss/wastage audit, settlements.	Complete control over workshop metal inventory.
9. Team & Multi-Branch	Staff management, granular 7-module RBAC, branch allocations, privilege escalation guards.	Multi-store management with strict security.
10. UI, Mobile & PWA	Multi-breakpoint responsive layouts, off-canvas navigation drawer, service worker, offline fallback.	App-like mobile & tablet experience everywhere.

1. Technical Architecture & System Foundations

JewelDesk is designed according to clean, modular Django architecture principles. Each functional business domain is encapsulated in an isolated Django application with dedicated models, views, forms, URL namespaces, and template sets.

Component	Technology / Pattern	Role in JewelDesk
Backend Framework	Django 6.1.1 (Python 3.12)	Robust MVC/MVT architecture, ORM, authentication, CSRF security.
Frontend & UI	Bootstrap 5.3.3 + Custom CSS & JS	Modern styling system, multi-theme tokens (Light, Dark, Gold).
Responsive Engine	Mobile-first Media Queries + Flex/Grid	Full compatibility across Phones (320px+), Tablets, Laptops & Desktops.
PWA Engine	W3C Manifest + Service Worker	App-like installability, caching shell, offline fallback page.
Document Engine	ReportLab PDF Library	Statutory GST tax invoices and enterprise operational reports.
Database	SQLite (Dev) / PostgreSQL (Prod)	ACID transaction safety, connection pooling & Supabase support.
Static Delivery	WhiteNoise Middleware	High performance, compressed, versioned static assets serving.

MULTI-THEME ARCHITECTURE:
JewelDesk supports three curated visual themes to suit showroom lighting conditions:

- **Light Theme:** Crisp white surfaces with subtle border dividers and dark typography. Optimized for well-lit office counters.
- **Dark Theme:** Deep obsidian background (#121824) with dark card surfaces. Eliminates glare in intimate jewellery showrooms.
- **Gold / Premium Theme:** Warm ivory and champagne surfaces with antique gold accents. Delivers a luxury jewellery brand feeling.

Theme persistence is implemented via cookie and localStorage sync, with an inline pre-paint script that guarantees zero theme-flicker on page load.

2. Detailed Module Specifications & Feature Breakdown

2.1 Executive Cockpit Dashboard

The Dashboard serves as the central command cockpit for the showroom proprietor and staff. It aggregates critical financial, inventory, and operational metrics in real time upon page load.

- **Today's Sales & Transactions:** Real-time revenue generated today with transaction counter.
- **Lifetime Revenue & Outstanding Due:** Total sales volume vs pending customer receivables.
- **Inventory Valuation:** Live retail stock valuation dynamically computed across all categories.
- **Low Stock Warnings:** Automatic design alerts when stock falls below pre-configured thresholds.
- **Pending Custom Orders:** Visual badge highlighting custom orders currently in production.
- **Quick Actions:** One-click shortcut tiles for New Sale, Add Customer, Inward Inventory, and Bespoke Order.
- **Live Metal Rates Widget:** Displays 24K Gold, 22K Gold, and 999 Silver prices with an API refresh button.

2.2 Inventory & Stock Management Module

Jewellery inventory requires exacting precision beyond standard retail goods. JewelDesk tracks every single piece by individual tag, hallmark, weights, and purity levels.

- **Item Identity & Tagging:** Unique Tag Number (barcode/RFID ready) and government-mandated HUID (6-character alphanumeric hallmark).
- **Metal & Purity Classification:** Gold (24K, 22K, 18K, 14K), Silver (999, 925 Sterling), and Platinum.

- **Three-Tier Weight Breakdown:** Gross Weight (total item weight), Stone Weight (diamonds, gemstones, pearls), and Net Weight (pure metal weight for billing).
- **Making Charges & Wastage:** Configurable making charge (per gram or fixed percentage) and wastage percentage.
- **Bulk CSV Inventory Import:** Upload hundreds of inventory records in seconds. Includes sample CSV download, preview confirmation, duplicate tag detection, and validation checks.
- **Audit Trail Stock Movements:** Every stock change creates an immutable StockMovement record (Inward, Outward, Sale deduction, Return addition, or Reconciliation Adjustment) logging user, timestamp, and reason.

2.3 Dynamic Pricing Engine & Metal Rates

Due to continuous fluctuations in bullion markets, JewelDesk decouples metal price updates from individual item records. When metal rates update, the entire catalogue re-prices dynamically.

- **Live Rate Integration:** Gold and silver rates can be fetched via live external financial APIs or manually updated by store managers.
- **Mathematical Pricing Model:**

JEWELDESK STANDARD PRICING FORMULA:

Metal Value = Net Metal Weight (g) × Current Metal Rate per Gram (for specific Karat)

Making Cost = (Making Rate per Gram × Gross Weight) OR (Metal Value × Making %)

Stone Value = Sum of (Stone Carats/Weight × Stone Rate)

Subtotal = Metal Value + Making Cost + Stone Value - Trade Discount

GST (3.0%) = Subtotal × 0.03 (CGST 1.5% + SGST 1.5% or IGST 3.0%)

Final Retail Invoice Price = Subtotal + GST

- **AJAX Pricing Calculator Endpoint:** `/pricing/api/calculate/` allows counter staff to calculate quotations instantly on the fly without page reloads.

2.4 Sales, Invoicing & Tax Billing Module

The sales module enables counter sales executives to generate statutory GST-compliant invoices in under 60 seconds.

- **Multi-Item Invoices:** Multiple jewellery items, coins, and loose stones on a single consolidated bill.
- **Tag Search & Quick Scan:** Scan or enter Tag Number to instantly populate metal weight, purity, HUID, and calculate prices automatically.
- **Customer Onboarding on the Fly:** Add a new customer directly inside the billing screen without losing cart state.
- **Multi-Mode & Split Payments:** Supports multiple payment tender types on a single invoice (e.g. ■50,000 Cash + ■75,000 Credit Card + ■25,000 UPI).
- **Partial Payments & Credit Ledger:** Record partial payments; the system automatically marks the invoice status as 'Partial' and tracks the pending balance.
- **ReportLab PDF Tax Invoices:** Generates professional A4 PDF tax invoices with store logo, GSTIN, HUID breakdown, tax breakup, payment receipts, and terms.
- **Instant WhatsApp Invoicing:** One-click WhatsApp button generates pre-formatted message links (<https://wa.me/>) to send invoices directly to customer phones.
- **Sales Analytics & Presets:** Preset filters for Today, This Week, This Month, and Custom Date Ranges with exportable totals.

2.5 Old Gold Exchange & Buyback Module

A cornerstone of the retail jewellery business is taking old scrap gold from customers. JewelDesk provides transparent calculation and auditable record-keeping.

- **Old Metal Karat Testing:** Staff record gross weight, stone weight deduction, and tested purity (e.g. 18K, 20K, 22K).
- **Melt Value & Melting Wastage:** Calculates pure gold content and applies permissible melting wastage percentage.
- **Two Operational Modes:**
 - *Exchange Mode:* Value of old gold is credited directly as a deduction on the customer's new purchase invoice.
 - *Buyback Mode:* Store purchases old gold for cash or bank payout with a signed statutory voucher.
- **Inward Scrap Audit Trail:** Detailed records of all old gold received, providing complete stock reconciliation when scrap is sent to refineries or karigars.

2.6 Customer Relationship Management (CRM)

JewelDesk CRM records long-term customer relationships, purchase ledgers, and sales leads.

- **Customer Profiles:** Full name, mobile number, email address, physical address, and anniversary/birthday for festive loyalty campaigns.
- **Statutory Compliance (KYC):** Fields for PAN card and Aadhaar numbers to satisfy mandatory compliance for cash sales over ■2,00,000.
- **Customer Financial Ledger:** Complete history of all lifetime purchases, invoices, returned items, and outstanding credit dues.
- **Enquiries & Lead Pipeline:** Record customer enquiries (e.g. bridal set enquiry, solitaire ring), scheduled follow-up dates, notes, and conversion status.

2.7 Custom Orders & Bespoke Jewellery Module

High-margin bespoke jewellery orders require strict milestone tracking from design to delivery.

- **Design Uploads:** Customer reference sketches, CAD renderings, and photo uploads stored securely.
- **Detailed Specifications:** Target weight, metal karat, gemstone details, sizing, engraving instructions.

- **8-Stage Lifecycle Tracking:** Pending → In Design → Approved → With Karigar → In Finishing → Ready for Delivery → Delivered → Cancelled.
- **Advance Payments & Quotations:** Tracks estimated order cost, advance deposit received, and remaining balance due on handover.

2.8 Karigar & Manufacturing Management Module

JewelDesk includes an integrated manufacturing submodule for tracking independent artisans and workshops.

- **Artisan Profiles:** Workshop names, contact details, specialized craft (Bengali, Kundan, Casting, Stone setting), making wage rates.
- **Raw Metal Issue & Work Orders:** Issue raw pure metal (gold/silver bars, grains) to karigars with recorded issue weight, purity, and expected delivery date.
- **Wastage & Metal Loss Auditing:** Compares issued weight against finished jewellery weight and returned scrap. Automatically flags excess metal loss exceeding allowable contractual wastage limits.
- **Karigar Settlements:** Tracks making wage payables, advances given, deductions, and settles accounts with payment vouchers.
- **Manufacturing Cockpit:** Real-time dashboard showing active workshop assignments, overdue jobs, and total metal currently held by artisans.

2.9 Team, Roles & Multi-Branch HR Module

JewelDesk provides enterprise-level security and multi-branch showroom control.

- **Employee Profiles:** Employee code, full name, designation (Store Manager, Salesperson, Cashier, Appraiser), contact info, login credentials.
- **Granular Role-Based Access Control (RBAC):** 7 distinct modules (Sales, Inventory, Pricing, Customers, Custom Orders, Karigar, Team) with independent View, Add, Edit, Delete flags.
- **Multi-Branch Store Support:** Group retail showrooms into branches (e.g. Main Showroom, Mall Branch). Staff and inventory can be mapped to specific branches.
- **Privilege Escalation Protection:** Strict security prevents regular employees from modifying their own permissions or assigning Administrator roles.

2.10 Responsive UI & Progressive Web App (PWA)

JewelDesk is designed to operate seamlessly across desktops, laptops, tablets (e.g. iPad at counter), and mobile phones.

- **Multi-Breakpoint Responsive Design:** Desktop (1200px+), Laptop (992px–1199px), Tablet (768px–991px), and Mobile (320px–576px).
- **Adaptive Off-Canvas Navigation:** Pinned 260px desktop sidebar transforms into a slide-out drawer on tablets and mobile with backdrop dismissal and body scroll-lock.
- **Zero Page Blowout:** Strict viewport safety ensures horizontal scroll containment inside table components, preventing annoying horizontal webpage scrolling.
- **Touch Target Accessibility:** Inputs enforce 16px font size on mobile to prevent iOS Safari auto-zoom, and buttons maintain ≥40px touch targets.
- **Web App Manifest (manifest.json):** Standard standalone PWA definition with custom luxury jewel icons (192px, 512px, maskable, apple-touch-icon).
- **Root-Scoped Service Worker (sw.js):** Network-first caching for dynamic business data, stale-while-revalidate for static assets, and pre-cached shell.
- **Dedicated Offline Fallback Page (offline.html):** Branded offline screen with connection detection, retry triggers, and automatic reconnection when internet restores.
- **In-App PWA Install Prompt:** Browser-native install banner and sidebar install button allowing store owners to install JewelDesk as a desktop or mobile application.

3. Step-by-Step Operational Workflows (Kese Kaam Krte H)

3.1 Workflow: Billing a New Jewellery Sale with Old Gold Exchange

Workflow Step	Action & Operational Description
Step 1: Open Sales Billing	Navigate to Sales & Exchange → New Sale (or click the 'New Sale' shortcut on the Dashboard).
Step 2: Customer Selection	Select an existing customer from the dropdown, or click 'Add Customer' to register a new customer in a lightweight modal.
Step 3: Add Items to Invoice	Enter or scan the Item Tag Number. JewelDesk instantly loads the item's Gross Weight, Net Weight, Purity, HUID, and calculates the retail price based on the live metal rate.
Step 4: Old Gold Exchange (Optional)	If the customer brings old scrap jewellery: click 'Add Old Gold Exchange'. Enter tested karat, gross weight, and melting wastage. The calculated value is automatically deducted from the bill total.
Step 5: Record Payment Tender	Enter payment modes (Cash, UPI, Credit Card, or Split). JewelDesk calculates balance due and marks payment status (Paid / Partial).
Step 6: Generate Invoice & Share	Submit the sale. Instantly print the ReportLab PDF tax invoice or click the WhatsApp button to send a digital bill directly to the customer's phone.

3.2 Workflow: Custom Order to Karigar Manufacturing & Handover

Workflow Step	Action & Operational Description
Step 1: Consultation & Booking	Navigate to Custom Orders → New Custom Order. Enter customer details, design notes, upload reference sketch/photos, specify target metal weight & karat, and collect advance deposit.
Step 2: Assign to Karigar	Open Manufacturing & Karigars → Work Assignments → New Assignment. Select the Karigar, link the Custom Order, issue pure gold/silver raw metal, and record the issued metal weight.
Step 3: Manufacturing & Tracking	The job progress is monitored on the Karigar Dashboard. The system highlights pending and overdue artisan assignments automatically.
Step 4: Receive Finished Piece & Audit	When the artisan brings the finished piece: record finished weight, returned scrap metal, and dust loss. JewelDesk calculates the wastage percentage and alerts if it exceeds agreed limits.
Step 5: Stock / Direct Delivery	The bespoke piece is approved, HUID is recorded, and the order status is updated to 'Ready for Delivery'.
Step 6: Customer Handover & Balance	Customer inspects the piece, pays the remaining balance invoice, and the order is marked 'Delivered'.

3.3 Workflow: Inventory Inward, HUID Tagging & Dynamic Re-Pricing

Workflow Step	Action & Operational Description
Step 1: Inventory Entry / CSV Upload	Add single jewellery items via Inventory → Add Jewellery Stock or upload hundreds via Import Inventory CSV file.
Step 2: Tag & HUID Recording	Assign unique Tag Number (e.g. RNG-2026-001) and Bureau of Indian Standards (BIS) Hallmark Unique ID (HUID) for legal hallmarking compliance.
Step 3: Weight Specification	Input Gross Weight, Stone Weight (carats/grams), and Net Weight. Set making charges per gram or as a percentage.
Step 4: Dynamic Re-Pricing	When bullion market rates fluctuate, staff click 'Refresh Rates' in Pricing Engine. All inventory items automatically recalculate retail prices instantly.
Step 5: Physical Reconciliation	Perform periodic stock audits via Adjust Stock to log physical verification counts and adjustments.

3.4 Workflow: PWA Installation & Offline Operation

Workflow Step	Action & Operational Description
Step 1: App Installation	Open JewelDesk in Chrome, Edge, or Safari. Click the 'Install JewelDesk' button in the sidebar or the browser address bar install icon. The app installs to your desktop or mobile home screen as a standalone application.
Step 2: Offline Resilience	If the showroom Wi-Fi or internet connection drops: JewelDesk automatically intercepts navigation requests via its Service Worker.
Step 3: Offline Screen & Data Safety	Displays a dedicated, branded Offline page reassuring staff that existing data is safe, and provides a 'Retry Connection' button.
Step 4: Automatic Reconnection	As soon as Wi-Fi/mobile internet reconnects, the application detects the online event and automatically refreshes to resume full operations.

4. Master URL Endpoints & Permissions Directory

URL Pattern	HTTP	Module / View Function	Access Role
/	GET	inventory.views.dashboard	Authenticated Staff
/inventory/	GET	inventory.views.inventory_list	Inventory View
/inventory/add/	GET, POST	inventory.views.inventory_add	Inventory Add
/inventory/import/	GET, POST	inventory.views.inventory_import	Inventory Add
/pricing/	GET	inventory.views.pricing_calculator	Pricing View
/pricing/api/calculate/	POST	inventory.views.api_calculate_price	Authenticated
/sales/	GET	sales.views.sale_list	Sales View
/sales/add/	GET, POST	sales.views.sale_create	Sales Add
/sales/<id>/pdf/	GET	sales.views.sale_invoice_pdf	Sales View
/sales/report/	GET	sales.views.sales_report	Sales View
/old-gold/exchange/	GET, POST	sales.views.old_gold_exchange_create	Sales Add
/customers/	GET	customers.views.customer_list	Customers View
/custom-orders/	GET	custom_orders.views.custom_order_list	Orders View
/karigar/	GET	karigar.views.dashboard	Karigar View

/karigar/assignments/	GET	karigar.views.assignment_list	Karigar View
/team/	GET	team.views.employee_list	Team View
/accounts/company-settings/	GET, POST	accounts.views.company_settings	Store Admin
/manifest.json	GET	config.pwa_views.manifest_view	Public
/sw.js	GET	config.pwa_views.service_worker_view	Public
/offline/	GET	config.pwa_views.offline_view	Public

5. Deployment, Security & Production Readiness

PRODUCTION DEPLOYMENT CHECKLIST & ENVIRONMENT CONFIGURATION:

JewelDesk is designed to run in production with complete security and high availability.

- **Database Storage:** During local development, SQLite (db.sqlite3) is used. In production, configure the DATABASE_URL environment variable to point to PostgreSQL (Supabase, Neon, AWS RDS, or standard PostgreSQL). Supabase transaction pooler compatibility is pre-configured with DISABLE_SERVER_SIDE_CURSORS = True and SSL enforcement.
- **Static Assets & WhiteNoise:** Static assets are served via WhiteNoise with CompressedStaticFilesStorage for automatic gzip/brotli compression and long-term browser cache headers. Run python manage.py collectstatic --noinput during deployment.
- **Email Dispatch:** Password reset and system alerts support SMTP (SendGrid, Mailgun, Amazon SES, or Gmail) via EMAIL_HOST, EMAIL_PORT, EMAIL_HOST_USER, and EMAIL_HOST_PASSWORD.
- **Live Bullion Rate API:** Set JEWELDESK_METAL_PRICE_API_URL and JEWELDESK_METAL_PRICE_API_KEY to enable automated daily metal rates fetch.
- **Security Hardening:** In production set DEBUG = False, configure ALLOWED_HOSTS, set CSRF_COOKIE_SECURE = True, SESSION_COOKIE_SECURE = True, and enable HTTPS enforcement.

SUMMARY CONCLUSION:

JewelDesk is fully functional, robustly tested, and features a production-grade responsive UI and PWA architecture. All business calculations, inventory tracking, hallmark compliances, PDF invoice rendering, artisan metal auditing, and security barriers are verified and operational.

Generated on behalf of JewelDesk Retail Management System.